

ESLtopia — Test Scenario Coverage Checklist

Every scenario that should be covered, and its current automation status · v1.7 · 2026-07-06

Purpose	A complete inventory of test scenarios across the application — not just what's automated, so gaps are visible and can be prioritized deliberately.
Companion documents	Master Test Plan · Test Case Specification (detail for every AUTOMATED/PARTIAL item below)
Legend	AUTOMATED covered by the current suite PARTIAL some coverage, notable gap remains GAP not automated today

v1.7 revision: §10's cross-browser row is updated, not re-flipped (it was already AUTOMATED). A live CI run — done specifically to verify v1.6 before trusting it — failed 5 `school-teachers.spec.ts` cases on WebKit with a real "email rate limit exceeded" error: running that file's real-invite-email calls under three engines in one run exhausted Supabase's shared email rate-limit bucket. `registration.spec.ts` and `forgot-password.spec.ts` carry the same latent risk (real signup/recovery emails) and were preemptively excluded too. All three are now Chromium-only for cross-browser purposes, same as the pure-REST files and visual suite. Re-verified with a second live CI run: 304 passed, 0 failed.

v1.6 revision: §10 (Cross-Cutting): cross-browser coverage flips from GAP to AUTOMATED. Two new `playwright.config.ts` projects run the existing UI/API suite under Firefox and WebKit (pure-REST files and the visual suite excluded — see Test Plan §13). No new test IDs were created; this is the same 144 cases now executed under up to three engines. Verified clean: the one failure and one flake seen in an initial full cross-browser run were confirmed by isolated reruns to be pre-existing parallel-load flakiness, not real per-browser defects.

v1.5 revision: Closed the three high-priority items from `test-coverage-todo.md`. §6 (Classes Management): all three Classes-tab UI click-through rows flip from GAP to AUTOMATED (CLSUI-01-04). §10 (Cross-Cutting): the XSS/HTML-injection row flips from GAP to AUTOMATED (CLSUI-05/06, RECU-09). §1 (Authentication): the rate-limiting row flips from GAP to **PARTIAL**, not AUTOMATED — LOGIN-17 and FPW-08 now exercise a burst of repeated attempts and lock in the invariant that failed attempts never succeed and the server never 500s, but as observed 2026-07-06 no lockout or 429 occurs within that burst on either endpoint, so the underlying protection this scenario describes isn't actually confirmed to exist. See the Test Plan §11 risk note.

v1.4 revision: A doc-sync pass against the actual `tests/` directory found three things this checklist had missed: (1) the visual-regression suite (§10) was live in CI but unlisted here; (2) a signup anti-enumeration test (REG-11) existed in code but wasn't reflected in §1; (3) an RLS insert-forgery test (CLS-08, renumbered from further down the file) existed in code but wasn't reflected in §3/§6. All three are now AUTOMATED rows below. No new test-writing happened in this pass — this only corrects the checklist to match code that already existed.

v1.3 revision: Record-sync failure handling (§7) is now closed — RECU-08 simulates a failed save (a resolved Supabase-error response, not a network exception) and confirms both that the error is logged and that the write still doesn't persist, since logging isn't durability. Both scenario gaps the v1.1 code review surfaced are now automated.

v1.2 revision: The teacher-removal cascade/warning gap is now closed — SHT-14 and SHT-15 (§5) verify the cascade actually deletes a populated teacher's classes/records and that the confirm dialog discloses it. Record-sync failure handling (§7) remains an open gap.

v1.1 revision: A code review pass fixed 8 code-level issues (see the Test Case Specification's revision note) but did not add new automated tests, so it surfaced — rather than closed — two scenario gaps: the teacher-removal cascade/warning

1. Authentication & Account Lifecycle

Status	Scenario	Notes
AUTOMATED	Valid login logs the user in	LOGIN-01, LOGIN-07/08/09
AUTOMATED	Wrong password / unregistered email → generic, non-enumerating error	LOGIN-02/03/10/11
AUTOMATED	Client-side validation: invalid email format, empty email/password	LOGIN-04/05/06
AUTOMATED	Unconfirmed user cannot log in, with the same generic message	LOGIN-16
AUTOMATED	Signup happy path (teacher and school) through confirmation to logged-in	REG-01/02
AUTOMATED	Signup form validation: mismatched/short/missing fields	REG-03–06
AUTOMATED	Signup role is restricted server-side even if the client is bypassed	REG-10, ROLE-02
AUTOMATED	Signup with an already-registered email does not reveal that an account already exists (anti-enumeration)	REG-11 — existed in code, added to this checklist 2026-07-06
AUTOMATED	Weak password rejected on both signup and reset	REG-12, PWR-03/04
AUTOMATED	Forgot-password flow does not reveal whether an email is registered	FPW-01, FPW-04/05
AUTOMATED	Completing a password reset invalidates the old password and the link itself	PWR-01, PWR-05
AUTOMATED	Sign out ends the session and does not silently restore it on reload	LOGOUT-01/02
AUTOMATED	Pending / inactive accounts are blocked at login with a specific message	ROLE-04/05, SCHAT-06
PARTIAL	Email deliverability and content (confirmation, invite, reset mail)	Confirmation is simulated via the admin API, not by reading a real inbox — deliberate tradeoff (see Test Plan §11), but actual email rendering/copy is unverified.
PARTIAL	Rate limiting / brute-force protection on login and password-recovery endpoints	LOGIN-17 and FPW-08 now exercise a small burst (4–5) of repeated attempts and confirm the invariant that failed attempts never succeed and the server never 500s. But as observed 2026-07-06, no lockout or 429 occurred within that burst on either endpoint — the test documents current behavior, it does not confirm brute-force protection actually exists. See Test Plan §11.
GAP	Account lockout / suspicious-activity handling	No such feature exists in the app today; flag if added.

2. Session Management

Status	Scenario	Notes
AUTOMATED	Unauthenticated visit shows the login screen, not the app shell	NAV-01
AUTOMATED	A valid session persists across a full page reload	NAV-02
AUTOMATED	A session for a since-deactivated account is force-signed-out on reload	NAV-03
AUTOMATED	A revoked/expired refresh token cannot mint new access tokens	LOGIN-15, LOGOUT-04
PARTIAL	Access-token expiry mid-session (not just at the refresh-endpoint level)	Only the raw refresh-grant API path is tested; a real in-browser session sitting past token expiry while mid-action isn't simulated.
GAP	Multiple concurrent sessions/tabs for the same user	Not exercised.

3. Authorization, Roles & RLS

Status	Scenario	Notes
AUTOMATED	Each role (teacher/school/superadmin) sees only its intended tabs	ROLE-03, SCHAT-05, SADM-01
AUTOMATED	A user can never elevate their own role via direct REST calls	ROLE-11/12, PROF-03
AUTOMATED	A user cannot read, modify, or forge ownership of another user's profile, class, or records	PROF-04, SADM-08/10, CLS-07/08/09/13
AUTOMATED	Superadmin can read/update/delete across all users' data; regular roles cannot	SADM-04-15, CLS-15/16
AUTOMATED	JWT claims (role, email, sub, exp) are well-formed for every role	ROLE-06-09, ROLE-13
PARTIAL	School-scoped data isolation (school A cannot see school B's teachers)	SCHAT-11 covers this at the API level; gated behind the undeployed feature flag.
GAP	Row-level security across every future table added to the schema	Structural risk, not a specific gap — flag whenever a new table is introduced without an accompanying RLS test.

4. Superadmin — Admin Panel

Status	Scenario	Notes
AUTOMATED	Admin tab visible only to superadmin; user table renders with Delete buttons	SADM-01/02/03
AUTOMATED	Changing a user's role via the Admin panel dropdown persists to the DB	SADM-16
GAP	Deleting a user via the Admin panel's Delete button (UI click-through)	Deletion is only tested at the REST layer (SADM-12); no test clicks the actual Delete button and confirms the row disappears from the table.
GAP	Admin panel search/filter/pagination, if the user table grows large	Not applicable today at current user counts; revisit if the panel gains these controls.

5. School → Teacher Management Feature not yet deployed

Status	Scenario	Notes
AUTOMATED	School can invite (create), deactivate/reactivate, and remove a teacher	SCHT-01, 04, 07 — automatic once the flag is on
AUTOMATED	Cross-school authorization is enforced on every management action	SCHT-02, 03, 11
AUTOMATED	Pending/deactivated teachers cannot log in; can self-activate via RPC	SCHT-06, 10, 12
AUTOMATED	Edge function input validation (missing email, bad action, duplicate email)	SCHT-08, 09, 13
GAP	Employees panel UI click-through (invite form, deactivate button, roster list)	Only tab visibility (SCHT-05) is UI-driven; the actual invite/manage UI has no click-through test — everything else calls the edge functions directly.
GAP	Invited-teacher email content and the setup link (?setup=1) flow end-to-end	Not exercised via UI.
AUTOMATED	Removing a teacher actually cascades to delete their classes/records, and the confirm dialog accurately warns about it	SCHT-14, SCHT-15 — added 2026-07-03 to close a gap a code review surfaced: SCHT-07's "remove" happy-path test previously gave the teacher no classes/records to lose, so the cascade (classes.owner_id / records.owner_id both cascade from auth.users) was never actually exercised.

6. Classes Management

Status	Scenario	Notes
AUTOMATED	Class CRUD (create, list, rename, delete) at the data layer	CLS-01-04
AUTOMATED	Adding/removing students from a class's roster	CLS-05/06
AUTOMATED	Data isolation between teachers on read/delete; superadmin can see everything	CLS-07/09, 15/16
AUTOMATED	A teacher cannot forge another teacher's owner_id to insert a class under someone else's ownership	CLS-08 — existed in code, added to this checklist 2026-07-06; closes the write-side half of the isolation story that CLS-07/09 (read/delete) already covered
AUTOMATED	Deleting a class cascades to delete its records	CLS-14
AUTOMATED	Classes tab UI click-through: creating a class via the "add class" form	CLSUI-01
AUTOMATED	Classes tab UI click-through: adding/removing a student via the roster input	CLSUI-02/03
AUTOMATED	Classes tab UI click-through: deleting a class via its delete button	CLSUI-04
GAP	Duplicate student names within a class; very long/unicode class or student names	No boundary/edge-case input testing on these free-text fields.

7. Records — Attendance, Grades & Payment

Status	Scenario	Notes
AUTOMATED	Empty state when a teacher has no classes yet	RECUI-01
AUTOMATED	Selecting a class populates the student table	RECUI-02
AUTOMATED	Marking a student present/absent updates the row and the daily summary	RECUI-03
AUTOMATED	"All present" quick-fill action	RECUI-04
AUTOMATED	Setting a grade and toggling payment persist across a reload (debounced autosave)	RECUI-05/06
AUTOMATED	Records CRUD and cross-teacher isolation at the data layer	CLS-10-13
PARTIAL	Student profile modal: attendance stats and note persistence	RECUI-07 covers the top-level stats and notes field; the trimester breakdown (attendance/grade history per term, final-grade dropdown) is not covered — see below.
GAP	Trimester breakdown in the student profile: expanding a term, viewing its attendance/grade log, setting a final grade	Rich sub-feature of the profile modal with no test coverage at all.
GAP	Changing the "class date" / "month" pickers to view a different day/month's data	Every current test uses "today"/"this month" implicitly; switching dates is untested.
GAP	Concurrent edits to the same record from two sessions (last-write-wins behavior)	The debounced upsert has no defined conflict resolution test.
AUTOMATED	A failed record save is logged, and does not silently count as persisted	RECUI-08 — added 2026-07-03. Intercepts the debounced upsert with a simulated 500 (a resolved error response, not a thrown network exception, to exercise the actual if (error) branch) and confirms both that the console error fires and that a reload shows the change was never actually saved.

8. Worksheet Generator

Status	Scenario	Notes
AUTOMATED	Selecting a topic and generating a worksheet opens the PDF preview with correct topic/grade metadata	PDF-01
PARTIAL	Worksheet generation across all task types (match, true/false, fill-in, listen-and-circle, color-the-boxes)	Only the fillin type is exercised (via the "am / is / are" topic, chosen specifically because it's deterministic). The other four task-type renderers have no generation test.
GAP	Grade filter (dropdown that narrows the topic grid to one grade)	Not exercised — the suite always leaves "All grades" selected.
GAP	Exercise-type pills (Match vs. True/False) for topics that support both	Not exercised.
GAP	Exercise-count input bounds (min 5 / max 20, non-numeric input handling)	Not exercised.

GAP	Assigning a class/student to a worksheet and confirming Name/Class fields populate on the sheet	Current PDF-modal tests generate without selecting a class or student.
GAP	Background worksheet toolbar (the "Show answers" / "New set" / "Preview PDF" controls behind the modal, before it's opened)	Generation immediately opens the modal in every existing test; the pre-modal toolbar path is unexercised.

9. PDF Preview Modal

Status	Scenario	Notes
AUTOMATED	Answer-key toggle shows/hides answers correctly	PDF-02
AUTOMATED	"New set" regenerates content while keeping the modal open	PDF-03
AUTOMATED	Modal closes via its close button and via the Escape key	PDF-04/05
GAP	"Print" button triggers window.print() with the expected print-only CSS applied	Click handler exists; actual browser print behavior and the @media print rules are not verifiable via Playwright's normal DOM assertions.
GAP	"Download PDF" opens a new tab with the worksheet HTML and expected styling	Not exercised — involves a new-window/popup flow and print-triggered download that's awkward to assert headlessly.
GAP	Pop-up-blocked fallback (alert shown when window.open returns null)	Not exercised.
GAP	Modal scroll position resets to top when "New set" is clicked	Behavior exists in code (pdfScrollRef.scrollTo) but is unverified.

10. Cross-Cutting Concerns

Status	Scenario	Notes
AUTOMATED	Visual/pixel-diff regression on four key screens (login, dashboard, worksheet generator settings, PDF preview modal) at a fixed 1280×900 viewport	VIS-01-04, <code>tests/visual/visual-regression.spec.ts</code> — existed in code (merged via the visual-regression-tests branch) but was missing from this checklist until now. Baselines are generated on CI's ubuntu-latest runner; regenerate via the "Update Visual Regression Baselines" workflow after intentional UI changes.
AUTOMATED	Cross-browser coverage (Firefox, WebKit — desktop)	82 of 144 cases (9 of 15 spec files) executed under the <code>firefox / webkit</code> Playwright projects — excludes the pure-REST files, the visual-regression suite, and (as of 2026-07-06, after a real "email rate limit exceeded" CI failure) every file that sends a real Supabase email: <code>registration.spec.ts</code> , <code>forgot-password.spec.ts</code> , <code>school-teachers.spec.ts</code> . See Test Plan §13. Mobile browser emulation remains a GAP.
GAP	Responsive/mobile layout correctness at the 860px and 480px breakpoints	CSS exists for both; no viewport-sized visual assertions. Distinct from the new visual-regression suite above, which only covers one fixed desktop viewport (1280×900), not these breakpoints.
GAP	Accessibility: screen-reader labeling, full keyboard navigation, color contrast	Only incidental keyboard support (Escape-to-close) is covered.
GAP	Performance under load (many classes/students/records for one teacher)	No defined SLA for this ~10-user portfolio app; out of scope per Test Plan.

AUTOMATED	XSS/HTML-injection resilience in free-text fields (student names, class names, notes)	CLSUI-05 (class name), CLSUI-06 (student name), RECUI-09 (notes) — confirm React's default escaping holds by injecting <code><script></code> / <code></code> payloads and checking they render as literal text with no matching DOM element and no execution.
GAP	Network-failure handling (Supabase request timeouts, offline state)	No offline/error-banner UX exists to test today.
PARTIAL	Deploy verification (confirming a shipped change is actually live in production)	Done manually per Test Plan §11 (bundle-hash check before trusting prod-run results) — not itself an automated Playwright test.

11. Summary

Of the areas surveyed above, the following are the highest-value gaps to close next, in rough priority order:

1. Worksheet generation across the remaining four task types (match, true/false, listen-and-circle, color-the-boxes), not just fill-in.
2. Student profile trimester breakdown (attendance/grade history per term, final-grade entry).
3. Employees panel UI click-through, once the school-teachers feature is deployed and the edge-function tests are unblocked.
4. Generator settings (grade filter, exercise-type pills, exercise-count input bounds) and the remaining Medium-priority items in `test-coverage-todo.md`.

Resolved 2026-07-06 (cross-browser rate-limit fix): a live CI verification run surfaced a real "email rate limit exceeded" failure on WebKit (5 `school-teachers.spec.ts` cases) — three engines sending real invite emails in one run exhausted Supabase's shared rate limit. Fixed by also excluding `registration.spec.ts` and `forgot-password.spec.ts` (same real-email mechanism) from the Firefox/WebKit projects. Re-verified with a second live CI run: 304 passed, 0 failed.

Resolved 2026-07-06 (cross-browser): Firefox and WebKit coverage (§10) is now automated via two new `playwright.config.ts` projects running the existing suite (minus pure-REST files and visual regression). No new test IDs; verified clean against production with no genuine per-browser defects found.

Resolved 2026-07-06 (test-writing): all three former High-priority backlog items are now automated or documented: Classes tab UI click-through (§6, CLSUI-01–04), XSS/HTML-injection resilience (§10, CLSUI-05/06 + RECUI-09), and rate-limiting behavior on login/recover (§1, LOGIN-17/FPW-08 — flipped to PARTIAL rather than AUTOMATED, since the tests document that no such protection is currently observed rather than confirming one exists). See the Test Plan §11 risk note on that last point.

Resolved 2026-07-06 (doc sync): this checklist was also brought back in sync with the code — the visual-regression suite (§10, VIS-01–04) is now listed as AUTOMATED, and two test cases that existed in code but were undocumented (REG-11 anti-enumeration in §1, CLS-08 RLS-insert-forgery in §3/§6) are now reflected here.

Resolved 2026-07-03: both scenario gaps the v1.1 code review surfaced — teacher-removal cascade (§5, SCHT-14/15) and record-sync failure handling (§7, RECUI-08) — are now automated.